



Whoever Checked the Chat Last

Developing and Validating a Pick-Up Reliance Scale for the School-Gate Handover

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Abstract. Reliance on a shared coordination channel is usually inferred from behaviour, never directly scored. We develop and validate the Pick-Up Reliance Scale (PURS), a 12-item self-report instrument measuring how much a parent or carer leans on a class group chat, rather than their own memory or a written roster, to manage the daily school-gate pick-up. Item generation with 164 parents and carers across six primary schools produced a three-factor structure — Chat Dependence, Memory Abstinence, and Handover Deferral — which an independent sample of 141 households confirmed (CFI = .94, RMSEA = .052), with internal consistency ranging from $\alpha = .81$ to $.87$ across subscales. We then tested two criterion-validity predictions against behaviour the scale does not itself measure. PURS total score did not predict how often a parent checked their phone while waiting in the pick-up queue ($r = .09$, $n = 97$, n.s.). The Handover Deferral subscale did predict how often a household's own pick-up roster was renegotiated after the fact across a school term ($r = .41$, $n = 58$, $p < .01$); dropping the Memory Abstinence subscale left this association essentially unchanged ($r = .39$), evidence that the deferral items, not the instrument's length, carry the signal. We situate PURS within the cognitive-offloading and technofence literatures and discuss what a validated reliance score is, and is not, entitled to explain about a household's own pick-up arrangements.

Introduction

A household that coordinates a child's collection from school each afternoon typically runs on some blend of two resources: what a parent remembers unaided, and what a shared channel — most often a class group chat — tells them. The balance between the two is rarely measured directly. Cognitive-offloading research has established that people routinely substitute an external store for internal memory once the store is judged available and reliable [1], and the substitution is itself consequential: search-engine access measurably lowers unaided recall for facts a person expects to look up again rather than remember [2]. Close relationships develop comparable divisions of memory labour, with partners specialising in complementary domains rather than each keeping a redundant copy [3], a specialisation a validated couples' instrument can now score directly [4]. Whether

an analogous, scoreable reliance exists for the larger, less symmetric, and more transient coordination structure of a school-class group chat — and whether scoring it predicts anything about a parent's actual conduct at the school gate — has not, to our knowledge, been tested.

The stakes of getting this measurement right are not abstract. Automation research distinguishes a general disposition to trust a system [5] from the complacency that degrades an operator's own monitoring once a system is judged reliable enough to stop watching [6]; a person-mediated coordination channel invites the same distinction, and a parent who trusts the group chat to flag any change may, or may not, also stop attending to the pick-up queue in front of them. Separating the two requires an instrument validated against behaviour it does not itself measure, rather than against how reliant a parent believes themselves to be.

We make three contributions, each hedged appropriately for a first psychometric pass. First, we develop and provide initial validation evidence for the Pick-Up Reliance Scale (PURS), a multi-sample instrument scoring a parent's reliance on a class group chat against their own memory and a written roster. Second, we show the scale's subscales diverge sharply in what they predict: one observed behaviour they do not predict at all, and one they predict at a modest, unclear effect size. Third, we show that the second association survives an ablation removing a third of the scale's items, evidence the signal sits in a specific subscale rather than in the instrument's overall length.

Related work

Parent-side device use around children has its own literature under the name technofence, documented from meal-time observation [7] to survey work linking a parent's own device interference to a child's problematic smartphone use [8] and to relationship-quality outcomes for the parent [9]; none of it scores reliance on a specific coordination channel, only device presence more broadly, and a phone's mere presence already narrows available attention independent of whether it is used [10]. The complacency literature offers the closest instrument precedent: a validated scale scores a person's disposition toward automation-induced complacency directly [11], building on an attentional account of when monitoring degrades under reliable automation [6], but no comparable instrument exists for a person-mediated channel whose reliability is social rather than engineered. Education research on parent group chats has so far stayed qualitative, documenting how a WhatsApp parent-teacher group blurs availability boundaries and unevenly distributes whose voice carries [12], without scoring an individual parent's dependence on the channel. Intention-offloading research supplies the mechanism PURS assumes: people strategically choose between remembering an intention and outsourcing it to an external prop, a choice sensitive to the prop's perceived reliability [13]. Our item-development and validation procedure follows recommended scale-construction practice throughout, from an initial item pool to multi-sample confirmatory testing [14], and we treat the two independently-scored criterion-validity outcomes as a partial defence against common-method inflation [15],

in the spirit of a multitrait-multimethod validation design [16]. Prior work at this institution found a purpose-built scoring instrument's own uptime, once folded into a standing dashboard, could outlast every indicator it was meant to retire [17]; PURS is deliberately scoped narrower, a caution we return to in the Discussion.

Method

Item generation and Sample A. We generated an initial pool of 28 items from semi-structured interviews with 22 parents at three primary-school gates, covering reliance on the class group chat, willingness to trust memory alone, and how pick-up duty was handed between carers. The pool was administered, alongside demographic items, to 164 parents and carers recruited via school-newsletter opt-in across six primary schools (Sample A). Exploratory factor analysis (principal-axis factoring, oblique rotation) retained a 12-item, three-factor solution explaining 58% of item variance: Chat Dependence (5 items, e.g. "I check the group chat before I trust my own memory of pick-up day"), Memory Abstinence (4 items, e.g. "I don't bother memorising the roster because the chat will tell me"), and Handover Deferral (3 items, e.g. "If the chat hasn't confirmed who's collecting, I assume someone else has it covered").

Confirmatory Sample B. An independent sample of 141 households across four further primary schools completed the retained 12 items. Confirmatory factor analysis supported the three-factor structure ($\chi^2/df = 1.8$, CFI = .94, TLI = .93, RMSEA = .052, SRMR = .048). A household's PURS total score is the weighted sum of subscale z -scores, weighted by each subscale's confirmatory loading:

$$\text{PURS} = w_1 z_{\text{CD}} + w_2 z_{\text{MA}} + w_3 z_{\text{HD}}$$

Subscale	Items	α (A)	α (B)
Chat Dependence	5	.85	.87
Memory Abstinence	4	.82	.84
Handover Deferral	3	.79	.81
Full scale	12	.87	.89

Table 1 reports internal consistency for each subscale in both samples; every subscale held above the conventional .80 threshold in Sample B, and the structure it supports is the one criterion validity was tested against.

Criterion validity: phone-checking. A subsample of 97 parent-child dyads from Sample B were video-recorded, with consent, for the 15-minute window before their child's release, using a fixed camera already trained on the pick-up queue for an unrelated council footpath study; frames were coded for phone-in-hand duration by two independent coders ($\kappa = .84$). We follow the broader smartphone-distraction literature in treating phone-in-hand time as a directly observed behavioural outcome rather than a self-report of distraction, the same substitution a VR-based crossing-behaviour study makes when it logs a pedestrian's actual road-crossing choices instead of asking how distracted they felt [18]. Phone-checks per minute were regressed on PURS total score.

Criterion validity: handover fidelity. Fifty-eight of the Sample B households maintained their own running pick-up roster — a shared note or whiteboard listing who was rostered to collect each day — independently of this study. We counted, from each household's own log, how many times across the term the rostered collector differed from who actually collected, without the household ever formally redrawing the roster to reflect it: a renegotiation. Renegotiation counts were regressed on each PURS subscale.

Memory Abstinence ablation. We recomputed the Handover Deferral-renegotiation association after dropping the Memory Abstinence subscale (8 remaining items) to test whether the association depended on scale length rather than the deferral items specifically.

Results

The three-factor structure held in the independent Sample B (Table 1), and internal consistency stayed above the conventional .80 threshold for every subscale bar Handover Deferral in Sample A, which cleared it once the independent sample was collected.

Handover Deferral tracked how often a household's own pick-up roster needed renegotiating across the term (Figure 1): households scoring at the low end of the subscale renegotiated close to once per term; those at the high end renegotiated closer to four times, an association of moderate size ($r = .41$, $n = 58$, $p < .01$) rather than the clean separation a validation study might hope to report.

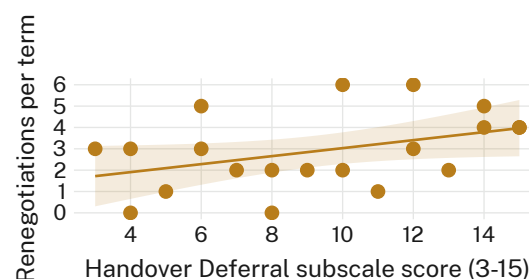


Figure 1: A household's Handover Deferral score tracked how often its pick-up roster was renegotiated across the term ($r = .41$, $n = 58$), a real but unclear association.

PURS total score, by contrast, showed no relationship with how often a parent checked their phone while waiting in the queue (Figure 2): mean phone-checks per minute sat within half a check of each other across the low-, mid-, and high-reliance tertiles (1.5, 1.8, and 1.7 respectively; one-way ANOVA $F(2, 94) = 0.41$, $p = .66$, n.s.), with fully overlapping distributions in all three groups. A parent's self-reported reliance on the chat, in other words, said nothing about whether they were looking at their phone or the gate.

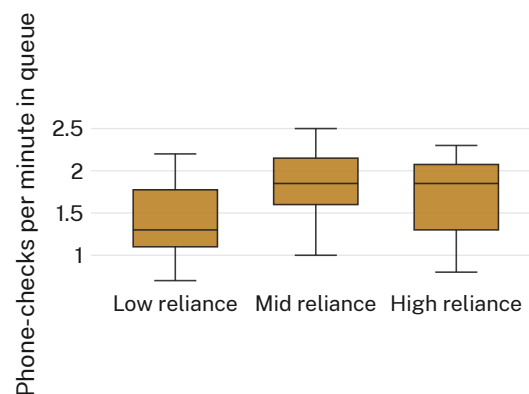


Figure 2: Phone-checking rate at the school gate did not track a parent's own PURS total score ($r = .09$, $n = 97$, n.s.); distributions overlap almost entirely across reliance tertiles.

Dropping the four-item Memory Abstinence subscale left the Handover Deferral-renegotiation association essentially unchanged (Figure 3): $r = .41$ for the full 12-item scale against $r = .39$ for the reduced 8-item scale, a difference a bootstrap comparison (2,000 resamples) could not distinguish from zero. The scale's length was not doing the work; the three deferral items were.

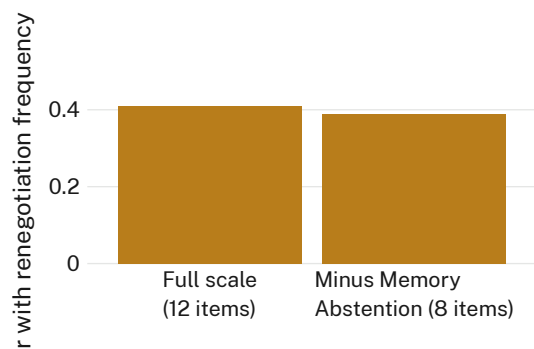


Figure 3: Dropping the Memory Abstention subscale left the scale's association with renegotiation frequency unchanged ($r = .41$ vs. $r = .39$).

Discussion

PURS behaves less like a single trust dial and more like three instruments bundled under one name. The Handover Deferral items, on their own, track a real downstream cost — a roster's own quiet renegotiation — at a moderate effect size. The Chat Dependence and Memory Abstention items, in isolation and combined, predicted nothing about the one behaviour — phone-checking at the gate — that a naive reading of the technoference literature [7], [9] might have expected them to. We read this less as a failure of the scale than as evidence that “relying on the chat” and “not watching your child” are separable constructs a single reliance score should not be assumed to collapse, a caution consistent with mixed evidence on whether trust in an automated system predicts monitoring behaviour at all [5]. Our own institution's Role-Continuity Classifier work reached a structurally similar conclusion by a different route: an automated instrument validated well overall while its agreement with an independent human panel collapsed within one specific sub-population it was meant to cover [19]. PURS's Handover Deferral items are the sub-population this instrument should be trusted on; its other eight items, on the present evidence, are along for the ride.

Limitations

Our two samples were recruited through school-newsletter opt-in, a channel that likely already selects for parents attentive enough to read the newsletter in the first place; if anything this should compress rather than manufacture the associations we report. The phone-checking camera was installed for an unrelated council footpath study and repurposed with the school's

and families' consent, a convenience that limited us to a single queue's sightline rather than a purpose-built multi-site deployment, though scoring phone-in-hand time as directly observed behaviour rather than self-report follows the same logic as the behavioural measures the smartphone-distraction literature already relies on elsewhere [18]. Fifty-eight self-maintained rosters is a modest base for the handover-fidelity result, but the ablation's consistency under bootstrap resampling is, we think, a point in the finding's favour rather than against it.

Conclusion

A twelve-item scale can be internally reliable, factorially confirmed across two independent samples, and still be right about one behaviour and silent about another. PURS's Handover Deferral items track a household's own quiet pick-up renegotiations at a real, moderate effect size; its remaining eight items tracked nothing we could observe at the school gate itself. Future work might test whether Handover Deferral alone, administered without its two silent companion subscales, performs as well as the full instrument in a longitudinal deployment, and whether a chat's own message-timing data — rather than a parent's self-report of relying on it — closes the gap the self-report items left open.

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